

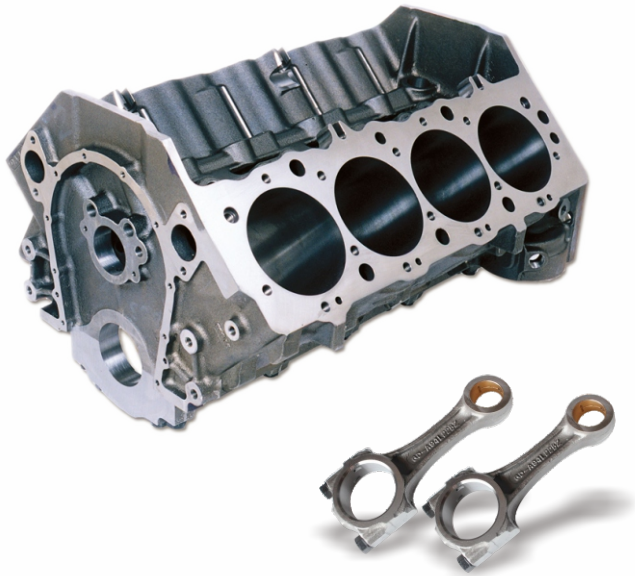
Tekintools honing tool series (split type)

- Honing head
- Honing bar
- Honing stone
- Inspection ring



Honing

Honing is a kind of precise and efficient production technique in machining operation. By honing, the size and precision of workpiece, shape and surface quality can be improved. Especially for machining inner hole of cylinder, crank hole, connection rod hole, oil pump, oil nozzle surface and other precise machining.



Tekintools has invested heavily in research and development of honing tool structure precision, expansion mechanism and auto measuring mechanism, then we lanchd a variety of honing products for different customer requests.

Careful manufacturing and material selection bring guarantee to product quality and show the Tekintools's ingenuity. excellent craft and technique make our honing tools be durable and precise.

Honing speed for different materials

Honing can bring high precision of size, shape and surface roughness. Tekintools honing tools can ensure IT6 precision, 0.003~0.005mm tolerance and Ra:0.40~0.10 roughness.

Material	Hardness	peripheral speed V m/s(m/mins)	Reciprocating speed Vf m/s(m/mins)
Steel	HB 200-300	0.42-0.97 (25-55)	0.33 (20)
	HB 230-475	0.30-0.75 (18-45)	0.30 (18)
	HRC 50-65	0.42-0.67 (25-40)	0.33 (18)
Cast iron	HB 200-475	0.83-1.08 (50-65)	0.42 (25)
	HRC 50-65	0.42-0.67 (25-40)	0.42 (25)
Aluminum	-	0.83-1.08 (50-65)	0.42 (25)
Cooper alloy	HB 120-140	0.83-1.08 (50-65)	0.42 (25)
	HB 180-200	0.75-0.92 (45-55)	0.42 (25)

Honing head

Single feed honing head

Single feed honing head is made up of honing head, expansion and contraction mandrel, honing base, honing stone, guide pad, auto measuring machanism, positioning spring. And guide pad is fixed on honing head, and honing stone is fixed on honing base.

Cylinder hole rough honing

Range: Φ 60- Φ 110

Economy honing head: Floating connection can offset coaxiality error between; workpiece and machine tool spindle; Auto measuring machanism can ensure tolerance 5-10 μ m;

According to cutting request, we can customize, improve and repair the honing tool;



Cylinder hole rough honing

Range: Φ 80- Φ 130

Floating connection;
Auto measuring machanism;
Several guide pads;
Cylindricity of cylider hole $\leq$ 0.008mm;
Size tolerance: $\pm$ 0.005mm;



Cylinder hole rough honing

Range: Φ 65- Φ 120

Floating connection;
Auto measuring mechanism;
Excellent stability;
According to cutting request, we can customize to design;
air measuring mechanism;



Double feeds honing head

Double feeds honing head is made of honing head, two sets of expansion mechanisms installed alternately, positioning spring, honing base and honing stone(Semi-finish), honing base and honing stone(Finish), measuring mechanism.

Cylinder hole rough honing

Range: $\Phi 60$ - $\Phi 110$

Excellent material selection, rigid treatment and careful manufacture;

Cylindricity $\leq 0.005\text{mm}$;

Size tolerance $\pm 0.003\text{mm}$;

According to cutting request, we can customize, improve and repair the honing tool;



Cylinder hole finish honing

Range: $\Phi 80$ - $\Phi 130$

Surface roughness can be $Ra0.40$ - 0.10 ;

High surface stability;

Increase workpiece life;

Decrease oil consumption and emission;



Cylinder hole finish honing

Range: $\Phi 60$ - $\Phi 100$

Double spherical connection can improve flexibility of floating mechanism
High precision Good cylindricity and roundness of hole.



Floating connection honing head
NEW

Honing bar

Honing bar is mainly for finish machining engine crank hole, gear hole, connection rod hold, oil pump hole and so on.

Tekintools honing tool cutting precision:

Crank hole coaxiality $\leq 0.015\text{mm}$, cylindricity $\leq 0.007\text{mm}$, roundness $\leq 0.006\text{mm}$

Connection rod, gear hole, oil pump cylindricity $\leq 0.007\text{mm}$, roundness $\leq 0.006\text{mm}$

Crank hole honing bar

Range: $\Phi 48$ - $\Phi 120$

Abrasive is made of diamond and possess sharp cutting edge and long tool life.

According to different workpiece material, we choose related abrasive.

There are guide pads on front and rear of honing bar, which can make precision higher.

Coaxiality $\leq \Phi 0.015\text{mm}$

Cylindricity $\leq 0.007\text{mm}$, roundness $\leq 0.006\text{mm}$



Connection rod honing bar

Range: $\Phi 45$ - $\Phi 60$

Good diameter conformity and cylindricity;

Easy to be used and suitable for batch processing;

High precision and efficiency with low cost;



Nozzle honing bar

Range: Φ5-Φ20

Honing bar is connected with spindle with rigidity connection, and floating connection between workpiece and holder;
Applied to honing that requires high straightness and adjustable diameter;
High cutting precision;
The tolerance of hold could be within 2-3μm, surface quality can achieve Ra0.2;



Small hole honing

Range: Φ4-Φ20

The production of small hole honing tool is a test both for maker and user. Tekintools can install delicate expansion mechanism and inner coolant supply mechanism, and we can ensure the precision of sizes and durability of tool. Honing bar has cutting abrasive and it's easy to be used.
Minimum machining holes≤Φ4
Size tolerance ≤0.0025mm, roundness≤0.003mm, cylindricity≤0.007mm
Roughness≤Ra0.2 Straightness≤0.001mm
According to cutting request, we can customize, improve and repair the honing tool.



For some materials to be machining hardly, such as rocker arm, gear, bearing and other special workpiece, it's a good choice to use honing to get high machining efficiency and cost down

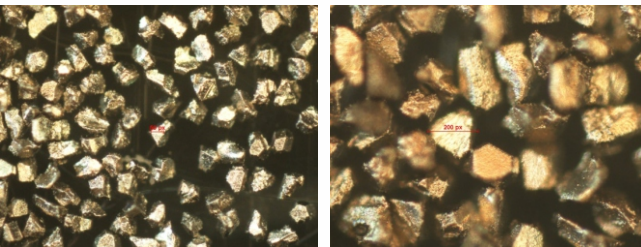
Honing stone

Honing stone is mainly used for finish honing, grinding, polishing and high precision machining. PCD and PCBN series are 300-500 times than traditional honing stone in tool life. Tekintools honing stone is more stable, and widely used for aerospace, automotive, internal combustion engine, air conditioning compressor and other industrial.



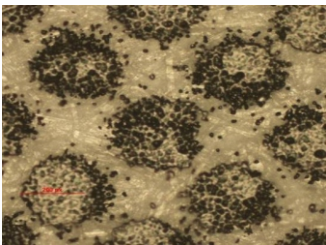
Abrasive:

White fused alumina(WA) is for honing carbon steel and alloy steel; Monocrystalline fused alunina (SA) or pink fused alumina(PA)is for honing stainless steel,bearing steel and high speed steel; PCBN is for honing stainless steel, high strength steel, superalloy and heat resistance steel; Carbide abrasive(GC, C, BC,D) is for honing brittle material.



Grain:

The grain has influence in roughness and productivity of machining workpiece surface. Thus we always choose the best grain of honing stone according to roughness and productivity.
80#-180# is for rough honing;
180#-280# is for semi finish honing; Finish honing is for above W40.



List of working pressure of honing stone on common material

Grain of honing stone	80	100-120	150	180	240	280	W40	W28	W20
Surface roughness R/μm	1.6-0.8	0.8-0.4	0.4	0.4-0.2	0.2-0.1	0.1	0.1-0.05	0.05	<0.05

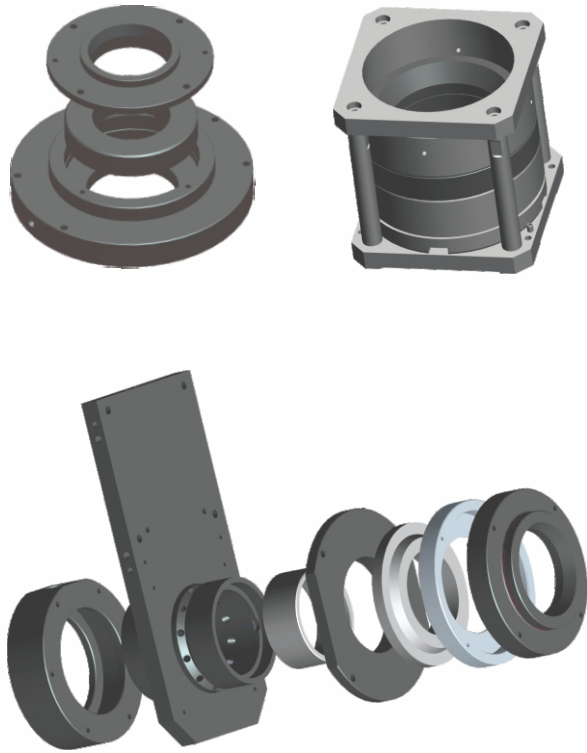
Inspection ring

Inspection ring is used for inspecting engine cylinder hole, and make online inspection for honing head and machine possible.

Inspection ring is a part of online inspection system. The quantity of inspection ring for honing per workpiece depends on process allocation allowance; Shape depends on clamping type of machine; Size of inner hole depends on each honing size of honing process.

Inspection ring generally divided into finish honing and rough honing. For inner hole size, rough honing is smaller than finish honing.

Common inspection ring



Production feature

By a large amount of tests of inspection ring material, Tekintools screened the best steel.

Heated in heat treatment furnace, bases got high hardness, high durability and good stability Using high precision machine to machining inner.

hole and outer radius, guarantees good precision, cylindricity, straightness and coaxiality.

Best
steel

High
durability

High
precision

Good
stability

Instruction

Before honing machining, the rough honing head must be adjusted with rough honing inspection ring; The finish honing head must be Instruction adjusted with finish inspection ring, then it can be used.

Notice

Using

When the inspection ring is installed at honing machine, it should be installed at the position easy to return to zero. If there is no any special change, the inspection ring should be installed at the setting position on machine. When the honing head needs returning to zero in inspection inner hole, it should go slowly for avoid breaking inspection ring.

Maintenance

There will be a lot greasy dirt on honing head after using, so it's necessary to clean it. Without maintenance and cleaning, much greasy dirt will lead to failure for returning to zero, therefore there is possibility that unqualified machining product exists or tool-crashing.

Regrinding

Normally, if there is wear in inner hole, for guaranteeing the accuracy of returning to zero, it's necessary to change new inspection ring in time instead of regrinding it.

Table for honing parameters of different materials

Workpiece material	Machining type	Cutting speed (m/min)	Cross Hatch Angle $\theta (^{\circ})$	Circumferential speed (m/min)	Reciprocating speed (m/min)
Cast iron	Rough machining	25-50	45-60	22-45	10-24
	Finish machining	-60	20-40	55-60	10-24
Steel	Rough machining	25-35	45-60	22-30	10-18
	Finish machining	-45	20-40	45-48	8-15
Carbide steel	Rough machining	30-40	45-60	28-37	12-20
	Finish machining	50	30	48	13
Chilled steel	Rough machining	20-25	45-60	18-23	8-13
	Finish machining	-35	30	34	9
Aluminium	Rough machining	30-60	45-60	26-55	12-24
	Finish machining	-70	20-40	65-70	12-24



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